Clear description of all columns

1. GRID_TYPE: The type of data table. In this file, it identifies the data as shot chart detail.
2. GAME_ID: Unique ID for the game.
3. GAME_EVENT_ID: Unique ID for the event within that game.
4. PLAYER_ID: Unique ID for the player who took the shot.
5. PLAYER_NAME: Name of the player who took the shot.
6. TEAM_ID: Unique ID for the player's team.
7. TEAM_NAME: Name of the player's team.
8. PERIOD: Quarter or overtime period when the shot happened.
9. MINUTES_REMAINING: Minutes left in the period at the time of the shot.
10. SECONDS_REMAINING: Seconds left in the minute at the time of the shot.
11. EVENT_TYPE: Result of the shot event, usually either Made Shot or Missed Shot.
12. ACTION_TYPE: The specific type of shot or move used, such as jump shot, layup, dunk, or step-back.
    Example: "ACTION_TYPE" tells you how the shot was taken.
13. SHOT_TYPE: Whether the shot was a 2PT Field Goal or 3PT Field Goal.
14. SHOT_ZONE_BASIC: General shot area, such as Restricted Area, Mid-Range, or Above the Break 3.
15. SHOT_ZONE_AREA: Side or section of the court where the shot was taken.
16. SHOT_ZONE_RANGE: Distance category of the shot.
17. SHOT_DISTANCE: Distance from the basket, measured in feet.
18. LOC_X: Horizontal court location of the shot.
19. LOC_Y: Vertical court location of the shot.
20. SHOT_ATTEMPTED_FLAG: Indicates whether a shot was attempted. In this dataset, it is always 1 because every row is a shot attempt.
21. SHOT_MADE_FLAG: Indicates whether the shot was made.
    1 = made, 0 = missed.
22. GAME_DATE: Date of the game in YYYYMMDD format.
23. HTM: Home team abbreviation.
24. VTM: Visiting team abbreviation

Statistical Data:

The dataset reveals that 2-point shots are attempted more frequently than 3-point shots (with success rates of 52.4% and 35.8%, respectively). Compared to 2-point shots, 3-point shots yield a higher expected point value per attempt. Furthermore, shots taken within the restricted area demonstrate the highest efficiency (a success rate of 63.5%), whereas mid-range shots are less efficient. Consequently—given that shots near the basket and 3-point shots (yielding 1.074 points per attempt) offer superior offensive value compared to mid-range shots (yielding 1.048 points per attempt)—shot selection exerts a profound influence on scoring efficiency.

Regression Analysis:

Logistic regression was employed to predict the likelihood of a shot being successful based on factors such as distance, shot type, quarter, and time remaining in the game. The coefficient for short distance was negative (-0.066, $p < 0.001$), indicating that the probability of a shot going in decreases as the distance increases. Each additional foot of distance results in a reduction in shot success probability (approximately 6.4%), assuming all other factors remain constant. This demonstrates that distance is a primary factor influencing the success of a shot.